

DSAP and PBAP Notes

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1 Using production-based asset pricing to discipline DSAP-implied discount rates

1.1 Two-sided equilibrium logic: demand and supply

Demand-system asset pricing (DSAP) and production-based asset pricing (PBAP) can be viewed as two mappings between the same equilibrium objects.

(i) DSAP (demand side). DSAP starts from investor portfolio demands and market clearing. Let p_t denote the vector of asset prices (or yields/spreads), and let $D_{h,t}(p_t, z_t)$ denote investor h 's demand as a function of prices and instruments/state variables z_t (e.g., regulation, index rules, relative supply shocks). Market clearing imposes

$$\sum_{h=1}^H D_{h,t}(p_t, z_t) = S_t, \quad (1)$$

where S_t is the (net) supply vector. Identification typically uses quasi-exogenous supply shifters (or relative-supply changes) to recover demand elasticities and infer equilibrium prices and risk premia.

(ii) PBAP (supply/production side). PBAP starts from firms' optimal investment policies given asset prices/discount rates. The firm chooses investment so that the marginal cost of installing capital equals the discounted marginal benefit (a first-order condition). PBAP therefore provides real-side restrictions linking discount rates to investment, capital accumulation, and Q .

1.2 DSAP-implied pricing objects

For the purpose of validation, suppose DSAP delivers either:

1. an *implied stochastic discount factor (SDF)* representation,

$$M_{t+1} = a_t - b'_t f_{t+1}, \quad (2)$$

where f_{t+1} are factor innovations (or state-price-driving shocks), and (a_t, b_t) are estimated from holdings/flows and supply instruments; or

2. an *implied expected excess return* (cost of equity/capital) for each firm/portfolio,

$$\hat{\mu}_{i,t}^e \equiv \hat{E}_t[R_{i,t+1} - R_{f,t}], \quad (3)$$

obtained from the estimated demand system under market clearing.

1.3 PBAP primitives: investment return and the investment Euler equation

Consider a firm with physical capital K_{it} , investment I_{it} , depreciation δ_{it} , and convex adjustment costs. Let κ denote the capital share in marginal profits (to avoid notation conflict with the adjustment-cost curvature parameter $a > 0$). A canonical specification is:

$$K_{i,t+1} = I_{it} + (1 - \delta_{it})K_{it}, \quad (4)$$

$$\Phi(I_{it}, K_{it}) = \frac{a}{2} \left(\frac{I_{it}}{K_{it}} \right)^2 K_{it}. \quad (5)$$

Let M_{t+1} denote the SDF relevant for the firm's shareholders. PBAP takes M_{t+1} as given and studies the implications of the firm's optimal investment.

Define the *investment return* as the ratio of the marginal benefit of installed capital at $t + 1$ to the marginal cost of investment at t :

$$r_{i,t+1}^I \equiv \frac{\text{marginal benefit at } t+1}{\text{marginal cost at } t}. \quad (6)$$

Under the benchmark (no corporate debt) setup, PBAP implies that the investment return equals the firm's stock return, realization-by-realization:

$$r_{i,t+1}^I = r_{i,t+1}^S. \quad (7)$$

The corresponding Euler equation is:

$$1 = {}_t[M_{t+1} r_{i,t+1}^I]. \quad (8)$$

Equation (8) is the key bridge: any candidate pricing kernel (including a DSAP-implied one) must price the investment return if firms are investing optimally.

1.4 PBAP as a long-horizon validation for DSAP

PBAP can discipline DSAP in two complementary ways, focusing on horizons over which capital adjustment is feasible (quarters to years).

A. Euler-equation pricing test (SDF consistency). If DSAP provides an implied SDF $\widehat{M}_{t+1}$ (e.g., via (2)), test the PBAP moment conditions:

$$\widehat{g}_i \equiv \frac{1}{T} \sum_{t=1}^T \widehat{M}_{t+1} r_{i,t+1}^I - 1, \quad \text{test } \widehat{g}_i \approx 0 \text{ for } i \in \mathcal{I}. \quad (9)$$

Here $r_{i,t+1}^I$ can be constructed from firm fundamentals using the PBAP expression for the marginal benefit and marginal cost, or approximated by stock returns under (7). Rejections can be interpreted as (i) misspecification of $\widehat{M}$, (ii) investment wedges (financing frictions, adjustment costs, measurement error), or (iii) failure of the benchmark PBAP assumptions.

B. Real-side pass-through test (cost of capital $\Rightarrow$ investment). If DSAP provides an implied time- t expected excess return $\widehat{\mu}_{i,t}^e$ (cost of equity), test whether it predicts subsequent investment in the direction implied by PBAP:

$$\frac{I_{i,t+h}}{K_{i,t+h}} = c_i + c_t + \psi \widehat{\mu}_{i,t}^e + \Gamma' X_{i,t} + u_{i,t+h}, \quad h \in \{1, 4, 8, 12\}, \quad (10)$$

where c_i and c_t are firm and time fixed effects, and $X_{i,t}$ controls for cash flow, sales-to-capital, leverage, and other standard investment covariates. A PBAP-consistent sign is typically $\psi < 0$: higher required returns (higher discount rates) should depress investment, holding expected marginal profitability fixed.

Interpretation. A key advantage of this validation design is that DSAP instruments that shift demand (relative supply, regulation, index reconstitutions) can generate plausibly exogenous variation in the cost of capital. PBAP then provides a disciplined way to ask whether these pricing movements propagate into real investment over longer horizons.